

Code-Layout, Readability and Reusability

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	4-space indentation used throughout notebook and app.py
2	Proper File Structure	Files and folders are logically organize	Yes	app.py, templates/, HDI.pkl, CSV file organized properly
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	df, X, y, model, y_pred, prediction used clearly
4	Function / Method Names	Functions are descriptively named	Yes	home(), predict() functions named descriptively in Flask
5	Code Comments	Inline and block comments explain log	Yes	Comments added in each cell of Jupyter notebook
6	Modular Design	Code is split into reusable functions/m	Yes	ML code in notebook, Flask routes in app.py separated
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code, clean and minimal codebase
8	Error Handling	Exceptions and errors are handled grand	Yes	Try-except used in Flask predict route

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	where Reused	Reusability Level (High / Medium / Low)
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1	HDI Prediction Model	Python / Scikit-learn	Flask app.py	High
2	Data Preprocessing Function	Python / Pandas	Jupyter Notebook	High
3	Flask Route Handler	Python / Flask	Web Application	Medium
4	Visualization Functions	Matplotlib / Seaborn	Jupyter Notebook	Medium
5	Pickle Model Loader	/ Pickle	Flask app.py	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized with clear folder structure
Readability	5	Clear variable and function names used
Reusability	4	Modular functions and reusable components
Documentation / Comments	4	Comments present in notebook and app.py
Overall Score	4.5	Excellent code quality throughout project